

FLEXTRADE FEATURE SPECIFICATION

Deviation Settlement Mechanism (DSM) Module

Regulatory Framework and Implementation Plan · Prepared 23 July 2026

1. Executive Summary

FlexTrade is adding a Deviation Settlement Mechanism (DSM) module to its platform: a feature that calculates, forecasts, and helps customers manage the financial exposure created when a participant's actual injection or drawal of electricity differs from its scheduled position under India's power market settlement framework.

Deviation exposure already sits at the center of FlexTrade's value proposition — Section 2 of the business model identifies deviation settlement minimization as the primary relevance of the platform for renewable developers, and Section 4 names generation forecasting as the core need for that segment. The DSM module operationalizes this by turning FlexTrade's existing forecasting and optimization capability into a transparent, block-level settlement calculator that customers can trust and act on before gate closure.

This document sets out the regulatory framework the module must implement, the product architecture, a phased implementation plan, and the metrics that will define success.

Why This Matters, By Segment

- RE Developers — direct reduction in deviation settlement penalties, the single largest controllable cost in a renewable PPA.
- BESS Operators — correct block-level deviation classification protects ancillary and arbitrage revenue from unrelated settlement charges.
- DISCOMs — portfolio-level deviation visibility across circles supports the procurement-cost reduction mandate in Section 4 of the business model.
- Energy Trading Companies — inter-state schedule deviations on DAM/RTM positions carry direct settlement risk that this module quantifies in real time.

2. Regulatory Framework

DSM in India has evolved through several regulatory cycles since it replaced the earlier Unscheduled Interchange (UI) mechanism. FlexTrade's module must be built as a configurable rule engine, not a fixed calculator, because the underlying regulation continues to change.

Year	Regulation	Key Change
2014	CERC (DSM and Related Matters) Regulations, 2014	Replaced the UI mechanism with a formal deviation settlement framework.
2015	First Amendment	Introduced a separate deviation charge and pricing structure for renewable generators.
2019	Fourth Amendment	Linked deviation charges to day-ahead market (DAM) prices and to the average grid frequency of the time-block.
2022	CERC (DSM and Related Matters) Regulations, 2022 (in force 5 Dec 2022)	De-linked deviation charges from grid frequency; charges instead based on time-block-wise prices from DAM, RTM, and the ancillary services market.
2024	CERC (DSM and Related Matters) Regulations, 2024 (notified 5 Aug 2024; effective date to be separately notified)	Replaces the 2022 regulations; introduces a “Normal Rate” (NR) of deviation charges based on the highest of the weighted average Area Clearing Price across DAM, RTM, or a combination, plus ancillary service charges.

Table 1 — Evolution of CERC's Deviation Settlement Mechanism regulations.

Current Settlement Logic

Under the framework FlexTrade must support, every 15-minute time block is settled against a Normal Rate (NR) derived from exchange-cleared prices, with the charge multiplier scaled by how far the entity's actual position deviates from its schedule — and, for renewable generators specifically, by how much they over-inject relative to schedule.

Party	Deviation Band	Illustrative Settlement Treatment
Wind / Solar Generator	Over-injection up to 5%	Paid at the Normal Rate — full settlement.
Wind / Solar Generator	Over-injection 5%–10%	Tariff for the excess reduced by 10%.
Wind / Solar Generator	Over-injection above 10%	Excess is not compensated.
Buyer / Drawee Entity	Over-drawal 10%–15%	Charged at 120% of the Normal Rate.
Buyer / Drawee Entity	Over-drawal above 15%	Charged at 150% of the Normal Rate.

Table 2 — Illustrative deviation treatment thresholds. Exact bands and multipliers must be validated against the CERC regulation version in force at deployment time.

Illustrative Deviation Charge Multiplier by Grid Frequency Band

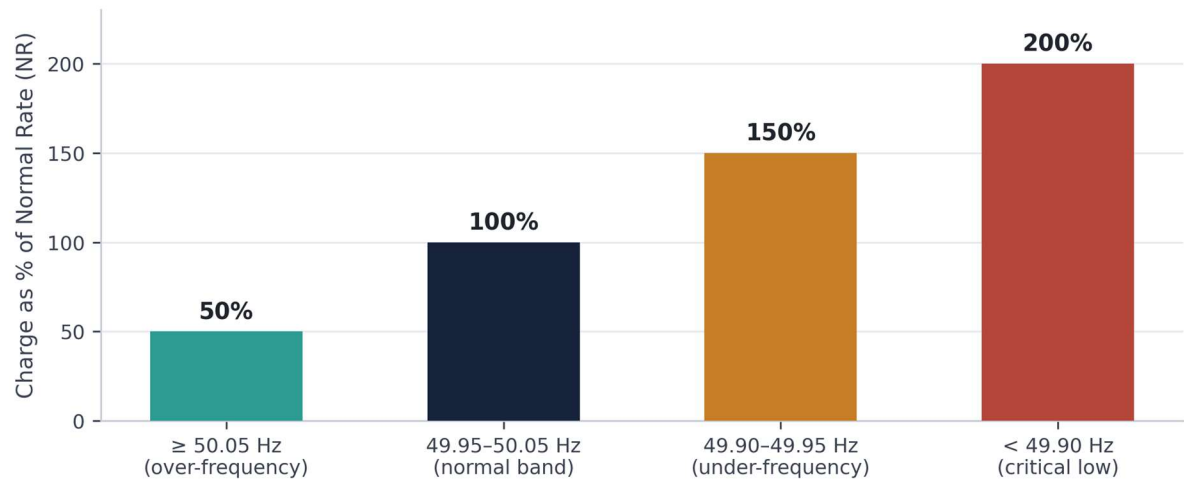


Figure 1 — Illustrative charge multiplier as grid frequency moves away from the 50 Hz reference, reflecting the frequency-sensitivity that DSM policy has historically built in around grid security events.

3. FlexTrade DSM Module — Product Framework

The module sits between FlexTrade's existing forecasting engine and the customer's settlement statements from the Regional Load Despatch Centre (RLDC), converting raw schedule-versus-actual data into an estimated, block-level DSM position before the official statement arrives.

Component	Function	Primary Data Source
Deviation Data Ingestion	Captures scheduled vs. actual injection/drawal at 15-minute block resolution.	SCADA, RLDC/SLDC schedules, exchange confirmations
Normal Rate Calculator	Computes the applicable NR from DAM/RTM/ancillary clearing prices per the regulation in force.	IEX / PXIL price feeds
Frequency & Band Classifier	Tags each time block against the applicable deviation band.	Grid frequency feed (NLDC / Grid-India)
Forecast Deviation Predictor	Forecasts expected deviation ahead of gate closure using FlexTrade's existing ML models.	FlexTrade forecasting engine
Settlement Reconciliation	Computes an estimated DSM charge or credit per entity per block and reconciles against actual RLDC statements.	RLDC settlement statements
Alerts & Revision Engine	Recommends schedule revisions to reduce exposure before the next gate closure.	Combines all of the above

Table 3 — Core components of the FlexTrade DSM module.

This module extends, rather than replaces, the forecasting and bidding features already described in the platform: the Forecast Deviation Predictor reuses the generation and demand forecasting models from the Renewable Scheduling and DAM/RTM workspaces, and the Alerts & Revision Engine surfaces its output through the same recommendation feed traders and RE developers already use.

4. Implementation Plan

The rollout is sequenced over 16 weeks, moving from regulatory mapping through a live pilot with reference customers before general availability.

Phase	Duration	Key Deliverables
1. Discovery & Regulatory Mapping	Weeks 1–3	Confirm the regulation version in force; map RLDC settlement data formats; define entity onboarding requirements.
2. Deviation Engine & Data Pipeline	Weeks 3–8	Build the ingestion pipeline, Normal Rate calculator, and frequency/band classifier.
3. Forecast & Alerting Integration	Weeks 7–11	Integrate with the existing forecasting engine; build the schedule-revision alert workflow.
4. Pilot with Reference Customers	Weeks 11–14	Run parallel settlement validation against actual RLDC statements with one RE developer and one DISCOM.
5. Rollout & Billing Reconciliation	Weeks 14–16	General availability; add module to Professional/Enterprise tiers; automate reconciliation reporting.

Table 4 — Phased implementation plan.

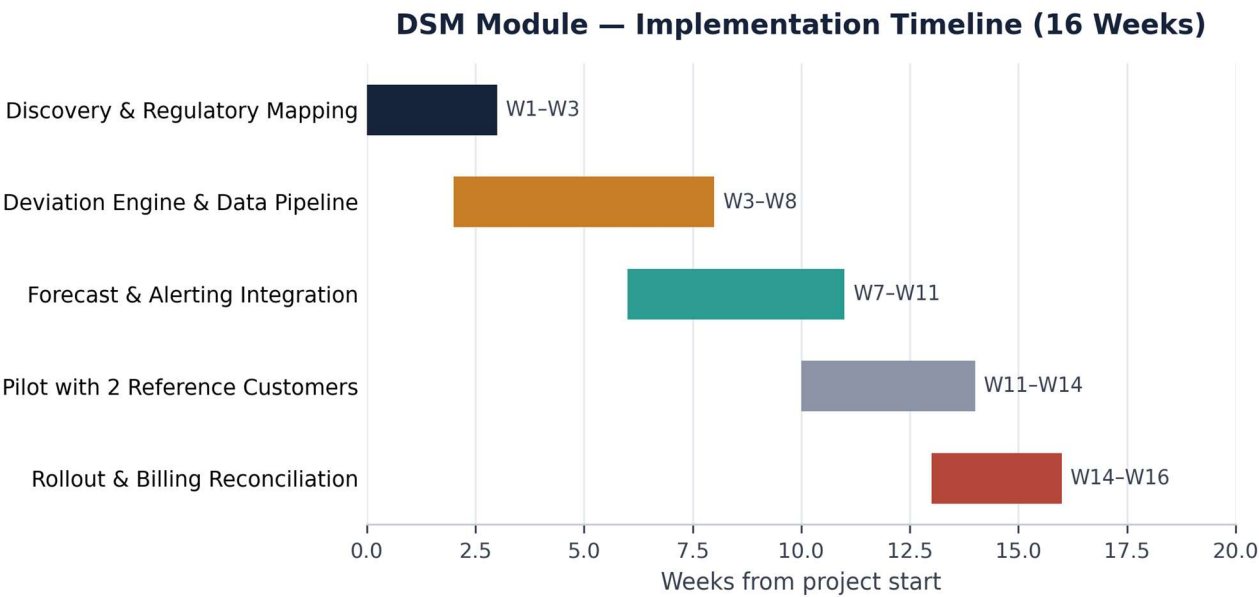


Figure 2 — Implementation timeline across the five phases.

5. Success Metrics

Metric	Target
Settlement calculation accuracy vs. RLDC statements	≥ 99%
Reduction in customer deviation penalty exposure	15–25% within two quarters of adoption
Alert lead time before gate closure	≥ 15 minutes
Pilot-to-GA customer retention	100% of pilot customers retained
Module adoption among Professional / Enterprise customers	60% within two quarters of GA

Table 5 — Success metrics and targets.

6. Risks & Mitigations

Risk	Mitigation
Regulatory version in flux (2022 vs. 2024 vs. future amendments)	Build the settlement engine as a configurable rule-set rather than hardcoded logic; monitor CERC notifications on an ongoing basis.
Data quality or latency from RLDC/SCADA feeds	Establish redundant feeds and reconciliation checks; flag stale data before it enters settlement calculations.
Customer trust in estimated vs. actual settlement	Publish the calculation methodology transparently; validate against actual RLDC statements throughout the pilot.
Cross-entity aggregation complexity (pooling / netting)	Model entity-level and pooled netting explicitly, per the applicable regulation.

Table 6 — Key implementation risks and mitigations.

7. Conclusion & Next Steps

The DSM module turns a regulatory cost center into a managed, forecastable line item for FlexTrade customers, and it deepens the platform's India-specific differentiation described in Section 7 of the business model — no adapted global platform can offer this level of settlement-mechanism fidelity out of the box.

Recommended next step: confirm sponsorship for Phase 1 discovery and begin regulatory mapping against the CERC (DSM and Related Matters) Regulations, 2024, ahead of its notified effective date.

Note: This document is a product planning framework, not legal or regulatory advice. Deviation charge bands, multipliers, and the Normal Rate methodology should be validated against the specific CERC regulation in force at the time of implementation.